

## Documentation

---

### COVID-19 Prevalence Forecasting App

Policy-Sensitive Scenario Forecasts (2020–2023)

---

### Table of Contents

- 1  Overview / Summary
  - 2  Data Sources & References
  - 3  Data Processing & Feature Engineering
  - 4  Model Architecture & Inductive Bias
  - 5  Temporal Structure & Forecast Horizons
  - 6  Spatial Coverage / Country Selection Logic
  - 7  Policy Modeling & Endogeneity Control
  - 8  Policy Effects & Causal Identification
  - 9  Visualization & User Interaction
  - 10  Version History, License, Evaluation & Caveats
- 

### 1 Overview / Summary

The **COVID-19 Prevalence Forecasting App** is a **research-oriented, retrospective forecasting system** designed to generate **probabilistic, uncertainty-aware projections** of **COVID-19 prevalence** across **time and countries**. It produces **multi-horizon probabilistic forecasts** of COVID-19 prevalence, expressed as **conditional quantile trajectories**, under **alternative policy as-if scenarios**.

The app is **explicitly not** a **real-time surveillance tool**, **early-warning system**, or **operational decision-support platform**. Instead, it is designed for **retrospective analysis** and **structured “as-if” scenario exploration** over the period **2020–2023**, with an emphasis on **transparency**, **interpretability**, and **clearly articulated epistemic limits**.

**Policy-sensitive forecasting** refers to a class of forecasting approaches in which **future trajectories** are explicitly **conditioned on alternative policy paths**, treated as **structured exogenous inputs** rather than as objects of **direct optimization** or **causal identification**. In this app, **policies** are represented as **normalized, time-indexed control paths** that enter the forecasting models as **scenario-defining conditions**.

These policy paths are **not interpreted as actionable levers** whose effects are assumed to be **known or causal**; rather, they define **counterfactual policy environments** under which forecasts are generated.

Rather than asking **what policy should be chosen**, policy-sensitive forecasting asks **how forecasts would plausibly differ** under **alternative policy trajectories** given the **same information constraints**, prioritizing **comparability** and **sensitivity analysis** over **prescription**. **Policy inputs** therefore serve as **scenario assumptions** rather than **estimates of policy effectiveness**.

**Forecasting**, in this context, refers to **retrospective, information-constrained projection**. For each **historical forecast origin**, the system conditions only on **information available at that time** and produces **multi-horizon probabilistic forecasts** forward. Instead of **single point estimates**, the system outputs **quantile-based forecast trajectories**, enabling structured reasoning about **uncertainty**, **dispersion**, and **upper and lower risk bounds**.

**Policy sensitivity** is implemented under **conservative, causality-favorable constraints**: **policy effects** are **bounded**, **horizon-aware**, and **monotonic**, limiting **extrapolation** and preventing **amplification of speculative effects**. This ensures that **scenario differences** reflect **structured plausibility** rather than **unconstrained counterfactual claims**, and that **uncertainty remains explicit** rather than collapsed into **point predictions**.

Taken together, the app is best understood as a **forecasting-based policy as-if scenario analysis tool** that emphasizes **transparency over optimization** and **explicit epistemic boundaries over causal claims**. Importantly, it can also be interpreted as a **best-case causal plausibility analysis** or **causal stress-testing framework**, providing an **upper bound on plausible policy leverage** under **favorable assumptions**, without asserting **causal identification** or **normative policy prescriptions**.

---

## **Data Sources & References**

### **Primary Data Source**

All **policy- and risk-related inputs** are derived from the:

#### **Cross-National COVID-19 Risk and Policy Dataset v1**

This dataset provides **harmonized, weekly, cross-national indicators** capturing **COVID-19 prevalence estimates** and **policy interventions**. It explicitly reflects the **reactive and endogenous nature of policy-making**, making it suitable for **forecasting contexts** where policy cannot be treated as **exogenous**.

### **Forecasting Data Artifacts**

The app uses **exactly two data artifacts**:

- **df\_final\_pre\_omicron.csv**
- **df\_final\_omicron.csv**

These represent **regime-specific transformations** of the source dataset. **No additional external policy datasets** are introduced at the forecasting stage.

---

### **Data Processing & Feature Engineering**

All **data processing** in the COVID-19 Prevalence Forecasting App builds directly on the **Cross-National COVID-19 Risk & Policy Dataset (v1)**. Only those **dataset components** required for **retrospective prevalence forecasting** and **policy-sensitive scenario analysis** are used; the app does **not** expose or depend on the dataset's **full variable inventory**.

#### **Temporal Alignment and Indexing**

All inputs are aligned to a **weekly global time axis**, corresponding to the dataset's **standardized Sunday-based reporting cadence**. A **shared global week index** ensures **consistent alignment** of **historical inputs** and **forecast horizons** across **countries and regimes**. **Feature construction** at each **forecast origin** uses only information that would have been **available up to that point in time**. **No forward-looking aggregation, smoothing, or normalization** is applied, ensuring **strict prevention of temporal leakage**.

The app operates on the dataset's **effective analytical window** from **March 2020 to July 2023**, corresponding to the period of **stable global COVID-19 reporting and policy activity**.

#### **Epidemiological Signal**

The **forecasting target** is **COVID-19 point prevalence**, defined as the **estimated share of the population actively infected** at a given time. **Prevalence estimates** are taken directly from the dataset's **harmonized prevalence construct**, derived from **surveillance and testing information** and adjusted for **under-ascertainment**. The app treats prevalence as a **noisy observational signal** rather than a **latent epidemic state** and does **not** impose **compartmental or mechanistic constraints**.

Prevalence values are **bounded in [0,1]** and transformed only as needed for **numerical stability** during model training. The **original scale** is preserved for **interpretation and visualization**.

#### **Policy Inputs**

**Policy-related inputs** are derived from the dataset's **harmonized government response indicators**. The app uses **normalized, continuous policy signals** representing **broad policy intensity families** rather than **individual legal instruments**. **Policy values** lie in the interval  $[0,1]$ , reflecting **relative intensity** within the observed data range.

Policy variables are treated as **endogenous and reactive by design**. They are **not residualized, detrended, or orthogonalized** with respect to epidemiological conditions. Instead, their influence is controlled **downstream** through **architectural constraints** and **lag gating**. As **policy activity declined globally**, policy inputs are **tailed toward zero** in the late period, consistent with the dataset's **documented handling of policy fade-out**.

### Synthetic Country Augmentation (Training Only)

In addition to dataset-based preprocessing, the **training pipeline** applies **synthetic country augmentation** as an **app-level strategy**. During **training only**, synthetic country trajectories are generated via **controlled recombination and rescaling** of observed country-level time series. The purpose is **not** to simulate real geopolitical entities, but to **expand trajectory diversity, reduce overfitting**, and encourage learning of **structural temporal dynamics** rather than **memorization**.

Synthetic augmentation is applied **exclusively during training**. **No synthetic data** are used for **validation, evaluation, or inference**. All forecasts produced by the app are based **solely on real historical country data**.

### Regime-Specific Training Artifacts

To accommodate **structural changes** associated with the **emergence of the Omicron variant**, the dataset is partitioned into **pre-Omicron** and **Omicron-era** training artifacts using the dataset's **documented regime definitions**. Both artifacts are processed using **identical logic** and are used **only during model training**. At inference time, the app relies exclusively on **observed historical data** and does **not require explicit regime labels**.

### Scope and Non-Goals

The **data processing pipeline** is designed to support **retrospective probabilistic forecasting** and **policy-sensitive scenario analysis**, **not causal identification** or **mechanistic epidemic modeling**. It does **not** attempt to infer **latent transmission parameters**, correct **all reporting biases**, or render **policy variables exogenous**. These limitations are **deliberate** and align with the app's emphasis on **stability, transparency, and explicit epistemic boundaries**.



## Model Architecture & Inductive Bias

## Architectural Philosophy

The **forecasting model** is designed to balance **nonlinear expressive capacity** with **explicit structural constraints** informed by **epidemiological reasoning** and the realities of **endogenous policy-making**. Rather than maximizing **flexibility** or **representational power**, the architecture deliberately introduces **inductive bias** to **stabilize learning** under **noisy, reactive inputs** and to prevent **implausible policy leverage**.

This design choice reflects a preference for **conservative, interpretable behavior** over **unconstrained fit**, particularly in a **retrospective, policy-sensitive forecasting setting**.

The model is **not intended** to discover **causal structure** or to simulate **epidemic mechanisms**. Instead, it is constructed to learn **stable, horizon-dependent statistical regularities** that are robust to **feedback between epidemiological conditions and policy responses**.

## Core Model Structure

The forecasting system is implemented as a **single global, horizon-aware neural quantile regressor** trained jointly across **countries** and **forecast horizons**. **Forecast horizon** is treated as an **explicit conditioning variable**, allowing the model to learn **horizon-specific dynamics** directly rather than extrapolating long-term behavior from short-term patterns.

All countries share a **common representation**; **no country-specific embeddings, fixed effects, or hierarchical random effects** are introduced. Countries are instead treated as **realizations of shared epidemic processes** observed under **heterogeneous conditions**.

**Historical prevalence signals, policy features, and horizon indicators** are mapped through a **shared nonlinear backbone** into a **latent representation**. From this shared representation, **separate quantile-specific output heads** produce **probabilistic forecasts** at multiple quantile levels. **Quantile crossing** is not formally prevented by hard constraints; instead, **coherence across quantiles** is encouraged implicitly through **shared representation learning** and **structural regularization**. Empirically, quantile crossing is **rare** under the chosen architecture and constraints.

## Constraints, Gating, and Capacity Control

**Policy-related inputs** are subject to **explicit structural controls** designed to mitigate **endogeneity** and prevent **implausible contemporaneous effects**. These include **lag-gated policy influence, directional monotonicity constraints, and explicit magnitude capping**.

During training, **policy gating** is applied **stochastically** to improve robustness and discourage reliance on **spurious associations**; during **validation and inference**, gating is **fixed conservatively**, enforcing disciplined behavior. **Effect magnitudes** are bounded through **policy-family-level caps**, with cap strength treated as a **tunable hyperparameter** optimized via **cross-validation**.

These mechanisms do **not** identify **causal effects** and are **not intended** to approximate **intervention response functions**. Instead, they act as **architectural regularizers** that bias the learned function class toward **conservative, causality-favorable behavior** while preserving **forecasting flexibility**.

### Target Variable and Modeling Scope

The **target variable, covid\_19\_prevalence**, is defined as the **estimated share of the population infected** and is **bounded in the interval [0,1]** by construction. **No explicit epidemic state-transition model, compartmental structure, or causal graph** is imposed. All **temporal structure** emerges from **learned statistical regularities** conditioned on **historical inputs** and **forecast horizon**.

Overall, the architecture should be understood as a **disciplined global forecasting model** that prioritizes **stability, plausibility**, and **explicit epistemic boundaries** over **mechanistic realism** or **causal identification**.

### Training and Inference Separation

All substantive **modeling assumptions**—such as **horizon-aware structure**, **constrained policy influence**, and **regularization choices**—are implemented during **model training** and are reflected in the resulting **trained model artifacts**.

At **inference time**, the application loads these **pre-trained artifacts** and applies them directly to generate **probabilistic forecasts** under **user-defined scenarios**. **No additional constraint enforcement** or **assumption checking** is performed during runtime. Consequently, the interpretation of all forecast outputs relies on the **documented training procedure** and the **provenance of the model artifacts** used.

---

## Temporal Structure & Forecast Horizons

The forecasting task is formulated as a **multi-horizon prediction problem**, in which **probabilistic forecasts** are produced simultaneously for a **fixed set of future horizons** relative to a given **historical forecast origin**. **Horizon** is treated as an **explicit modeling dimension** rather than as a by-product of **recursive extrapolation**.

**Forecast horizons** are defined on the same **weekly time scale** as the input data and extend forward from each **forecast origin** over a **fixed maximum horizon**. For each

horizon, the model outputs a **full set of conditional quantile forecasts**, allowing **uncertainty** to be represented **explicitly and separately** at each lead time.

**Near-term horizons** are driven primarily by **recent observed prevalence dynamics** and **short-run temporal regularities**, while **longer horizons** increasingly reflect **learned structural patterns** summarizing how prevalence trajectories tend to evolve under **similar historical and policy conditions**. **Uncertainty naturally increases with horizon length**, and this increase is **modeled explicitly** rather than suppressed through **smoothing** or **recursive certainty propagation**.

Crucially, **horizon dependence** is modeled directly. **Forecast horizon** enters the model as an **explicit conditioning variable**, allowing the learned mapping from **historical inputs** to **future outcomes** to vary systematically with **lead time**. This design avoids the accumulation of **artificial certainty** that can arise in **recursive or iterated forecasting approaches** and ensures that **long-horizon forecasts** remain **probabilistic, conservative, and interpretable**.

As a result, **forecast uncertainty, dispersion, and sensitivity to policy scenarios** are all **horizon-specific quantities**. Differences between scenarios should therefore be interpreted in a **horizon-aware manner**, with **greater epistemic caution** applied as **forecast lead time increases**.

---

## **Spatial Coverage / Country Selection Logic**

The forecasting app operates on a **cross-national panel** derived from the **source dataset**.

Countries are included based on the availability and consistency of:

- **prevalence estimates,**
- **policy indicators,**
- **sufficient temporal coverage.**

Countries are treated as **realizations of shared epidemic processes**, not as **independent** or **directly comparable experimental units**.

The app's spatial coverage corresponds **exactly** to the analytical sample of the underlying **Cross-National COVID-19 Risk & Policy Dataset (v1)**. The dataset includes **89 countries and territories**, selected based on **sustained availability** of **policy indicators, epidemiological risk measures, and sufficient temporal coverage** over the **2020–2023 period**.

Coverage is **global but intentionally non-universal**, prioritizing **cross-national comparability** and **data reliability** over **maximal geographic inclusion**. Countries are listed below by **broad region**, together with their **ISO-3 country codes**.



**Africa:**

Botswana (BWA), Namibia (NAM), Rwanda (RWA), Senegal (SEN).

**Asia:**

United Arab Emirates (ARE), Azerbaijan (AZE), Bahrain (BHR), Bhutan (BTN), Hong Kong (HKG), Indonesia (IDN), India (IND), Iran (IRN), Israel (ISR), Japan (JPN), Jordan (JOR), Kazakhstan (KAZ), South Korea (KOR), Kuwait (KWT), Sri Lanka (LKA), Malaysia (MYS), Philippines (PHL), Qatar (QAT), Saudi Arabia (SAU), Thailand (THA), Taiwan (TWN), Vietnam (VNM), Turkey (TUR).

**Europe:**

Albania (ALB), Andorra (AND), Austria (AUT), Belgium (BEL), Bulgaria (BGR), Belarus (BLR), Switzerland (CHE), Cyprus (CYP), Czechia (CZE), Germany (DEU), Denmark (DNK), Spain (ESP), Estonia (EST), Finland (FIN), France (FRA), Faroe Islands (FRO), United Kingdom (GBR), Georgia (GEO), Greece (GRC), Croatia (HRV), Hungary (HUN), Ireland (IRL), Iceland (ISL), Italy (ITA), Liechtenstein (LIE), Lithuania (LTU), Luxembourg (LUX), Latvia (LVA), Malta (MLT), Netherlands (NLD), Norway (NOR), Poland (POL), Portugal (PRT), Romania (ROU), Kosovo (XKX), Russia (RUS), Serbia (SRB), Slovakia (SVK), Slovenia (SVN), Sweden (SWE), Ukraine (UKR).

**Oceania:**

Australia (AUS), Fiji (FJI), New Zealand (NZL).

**North America (including the Caribbean):**

Bahamas (BHS), Barbados (BRB), Canada (CAN), Costa Rica (CRI), Cuba (CUB), Dominican Republic (DOM), Jamaica (JAM), Mexico (MEX), Panama (PAN), Trinidad and Tobago (TTO), United States (USA).

**South America:**

Argentina (ARG), Brazil (BRA), Chile (CHL), Colombia (COL), Ecuador (ECU), Peru (PER), Uruguay (URY).

All listed countries meet **minimum inclusion thresholds** for **policy reporting continuity, epidemiological signal availability, and institutional capacity**.

Consequently, outputs produced by the app should be interpreted as applying to this **medium- to high-capacity cross-national sample** and should **not** be generalized to countries outside this set without **additional validation** or **structural adjustment**.

---

## **Policy Modeling & Endogeneity Control**

### **Policy Endogeneity**

**Policy variables** are **endogenous by construction**: government interventions are typically enacted in response to **worsening epidemiological conditions** rather than independently of them. As a result, **naïve inclusion** of policy indicators risks learning



**spurious contemporaneous associations** driven by **reverse causality** or **shared shocks** rather than genuine policy influence.

The modeling strategy therefore treats policy inputs as **reactive signals** whose influence must be **structurally constrained** rather than statistically “controlled away.”

### **Lag Gate (Endogeneity Mitigation)**

To mitigate **policy–outcome feedback**, a **structural lag gate** is applied to **policy-related inputs**. This gate suppresses **direct contemporaneous policy effects** and allows policy influence to emerge only with **temporal separation** from the **forecast origin**.

During **training**, the lag gate is **randomly sampled** to improve robustness; during **validation and inference**, the gate is **fixed conservatively**, enforcing disciplined treatment of **policy timing**. This mechanism reduces sensitivity to **reactive policy changes** but does **not** identify **causal effects** or render policy inputs **exogenous**.

### **Directional Constraint & Effect Capping**

Additional **structural constraints** prevent **implausible or unbounded policy leverage**. **Smoothed policy features** are restricted from generating **systematic increases in projected prevalence** when active. **Policy effects** are also **capped in magnitude** to reflect **finite, saturating intervention influence**.

These constraints **do not establish causality**. Instead, they bias inference toward **causality-favorable, conservative behavior**, supporting **upper-bound reasoning** about plausible policy leverage rather than **effect estimation**.

### **Policy-Only Scenario Manipulation**

In **retrospective as-if scenario analyses**, only **policy-related inputs** are subject to manipulation. **Historical epidemiological trajectories** are treated as **fixed and observed**. The system therefore generates **conditional forecasts** under alternative policy trajectories, **not simulations** of fully counterfactual epidemic worlds.

---

## **Policy Effects & Causal Identification**

This app does **not estimate, identify, or validate causal effects** of policy interventions on COVID-19 prevalence. All policy-related inputs are treated as **endogenous, historically reactive signals** rather than **exogenous treatments**.

Scenario-based differences reflect **conditional projections**, not **formal counterfactual outcomes**. Structural constraints bias the system toward **causality-**

**favorable interpretations**, enabling **causal stress-testing** rather than **causal inference**.

Outputs should **not** be used to justify **policy prescriptions**, estimate **treatment effects**, or compare **intervention effectiveness** across countries.

---

## **Visualization & User Interaction**

The frontend is implemented using **Streamlit** and serves as a **lightweight, interactive interface** for inspecting **retrospective prevalence forecasts** under **alternative policy scenarios**. The interface emphasizes **exploration, comparison, and interpretability**, not **automation** or **decision support**.

Users select a **country**, a **historical forecast anchor week**, and define **forward-looking policy scenarios**. Forecast outputs are visualized as **median trajectories** with **explicit uncertainty bands** across horizons. All controls correspond **directly to model inputs**, and no hidden aggregation or recommendation logic is applied.

---

## **Version History, License, Evaluation & Caveats**

### **Version History**

#### **v1.0 — Initial public release**

- Probabilistic prevalence forecasting
- Quantile-based outputs
- Synthetic country augmentation
- Lag-gated policy endogeneity control
- Directional policy constraint
- Effect capping with hyperparameter optimization

### **Evaluation & Calibration**

Model evaluation focuses on **quantile loss, coverage behavior, and stability across horizons and regimes**. Uncertainty estimates reflect **model-based uncertainty only**.

### **License**

Released under the **MIT License** (unless otherwise specified).

### **Citation**

Christoph H. Maass. (2025). *COVID-19 Prevalence Forecasting App: Policy-Sensitive Scenario Forecasts*. GitHub repository:  
[https://github.com/chhmaass/covid\\_prevalence\\_forecasting\\_app](https://github.com/chhmaass/covid_prevalence_forecasting_app)

### ⚠️ Caveats & Interpretation Boundaries

This system produces **retrospective, conditional probabilistic projections** that:

- are **not** formally identified causal effects,
- do **not** imply policy prescriptions,
- are **not** validated for real-time surveillance,
- should be interpreted as **policy as-if scenario analysis** and **causal stress-testing**.

---

### 📄 Final Note

This documentation is intended to support **inspection, reasoning, and reuse**.  
It is written to make **assumptions and boundaries explicit**, not to **assert authority**.